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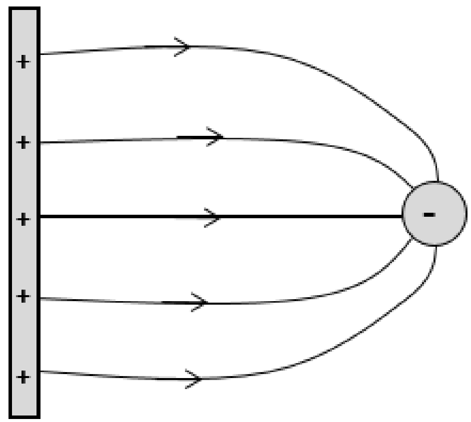
(a)	Electric field strength at a point is the electric force experienced per unit charge on a small stationary positive point test charge placed at that point and is in direction of the electric force on a positive test charge at that point.
(b)(i)	
	Correct direction of E field E field stronger to the right Symmetrical field pattern At least 5 lines
(b)(ii)1.	$F = Bqv \sin 90°$
	$= 3.0 \times 10^{-10}$ N
(b)(ii)2.	Direction of magnetic field due to wire is into the page and current is downwards. Using FLHR, direction of magnetic force experienced by the charge is to the right.
	Force has a different direction from velocity of charge, hence path is curved.
(b)(ii)3.	Apply a uniform electric field to the right OR using parallel plates with positive plate at the left
	Net force on charge will be zero due to electric force which is opposite in direction but equal in magnitude to the magnetic force exerted by the rod.
	Alternative Apply a magnetic field out of page OR by placing another rod carrying current of the same magnitude and direction such that the charge lies exactly between the 2 rods
	Net force on charge will be zero due to magnetic force which is opposite in direction but equal in magnitude to the magnetic force exerted by the rod.

(b)(iii)1.	$B = \frac{\mu_o I}{2\pi r}$
	= 4.0 x 10^{-9} T
(b)(iii)2.	Using RHGR, magnetic field due to coil at P is out of the page while magnetic field due to rod at P is into the page.
	Hence resultant magnetic flux density at P is the vector sum of the magnetic field due to the coil and rod.
(b)(iii)3.	The part of the coil closer to the rod will experience a repulsive force / force to the right (current in opposite direction) while the opposite end of the coil will experience an attractive force / force to the left (current in the same direction)
	The repulsive force is stronger than the attractive force as the magnetic flux density decreases with distance from the rod, therefore the coil will experience a net force to the right.
(c)(i)	The area under the graph between 200 ms and 270 ms represents the loss in magnetic flux linkage experienced by the coil when the magnet is leaving the coil.
(c)(ii)	As the magnet approaches the coil, the coil experiences a change of magnetic flux linkage and emf is induced, reaching a maximum when the N-pole of the magnet reaches the coil.
	When the S-pole leaves the coil, rate of change of magnetic flux linkage is larger as speed increases due to acceleration, resulting in a higher induced emf.
	As the change in magnetic flux linkage is in the opposite direction when the magnet is entering and leaving the coil, with increasing flux when entering and decreasing flux when leaving, induced emf have opposite polarity.
(c)(iii)	Due to greater speed attained, there is a greater rate of change of magnetic flux linkage experienced by the coil.
	The two peaks will be higher as magnitude of induced emf increases.
	OR
	Due to greater speed attained and change in magnetic flux linkage remains unchanged.
	The time duration where there is an induced emf is shorter.
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